C4 Problems: trigonometric functions

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Angles are in radians unless a degree sign is written. "Bereken exact" means: use exact values, no calculator.

Practice

1. Convert.

a. $135°$, $210°$ and $330°$ to radians (exactly).
b. $\frac{5\pi}{3}$ and $\frac{7\pi}{12}$ to degrees.
c. 1 radian to degrees, rounded to one decimal.

2. Give the exact values.

a. $\sin\frac{5\pi}{6}$ b. $\cos\frac{4\pi}{3}$ c. $\tan\left(-\frac{\pi}{4}\right)$ d. $\sin\frac{7\pi}{4}$ e. $\cos\frac{11\pi}{6}$ f. $\tan\frac{2\pi}{3}$

3. Given $f(x) = 3 + 2\sin\left(4\left(x - \frac{\pi}{8}\right)\right)$.

a. Give the equilibrium, amplitude, period and a starting point.
b. Give the maximum and minimum values and the first $x > 0$ where each occurs.
c. Sketch the graph on $[0, \pi]$.

4. Solve $\sin x = -\frac{1}{2}$. Give all solutions, and the solutions in $[0, 2\pi]$.

5. Solve exactly: $\cos 2x = -\frac{1}{2}\sqrt{3}$ on $[0, 2\pi]$.

6. Give all solutions.

a. $\tan x = \sqrt{3}$
b. $\tan 2x = -1$

7. A circle has radius 6 cm. A central angle of $\frac{2\pi}{3}$ cuts off an arc and a sector. Give the exact arc length and sector area.

8. Prove these identities.

a. $(\sin x + \cos x)^2 = 1 + \sin 2x$
b. $\dfrac{1 - \cos 2x}{\sin 2x} = \tan x$ (where $\sin 2x \neq 0$)

9. In triangle ABC: $\alpha = 40°$, $\beta = 65°$ and $a = 10$. Calculate γ, b and c (two decimals).

10. A triangle has sides 7, 8 and 13. Bereken exact the largest angle.

11. A sinusoid has a maximum at $(1, 7)$ and the next minimum at $(4, 1)$. Give a formula of the form $y = a + b\cos(c(x - d))$, and one of the form $y = a + b\sin(c(x - d))$.

CE-level

12. Solve exactly on $[0, 2\pi]$: $\sin x = \cos 2x$.

13. Solve exactly: $\sin 2x = \sin\left(x + \frac{\pi}{3}\right)$. Give all solutions, then those in $[0, 2\pi]$.

14. Bereken exact $\cos\frac{\pi}{12}$ using an addition formula.

15.

a. Toon aan that $\sin x + \sqrt{3}\cos x = 2\sin\left(x + \frac{\pi}{3}\right)$.
b. Give the maximum value of $\sin x + \sqrt{3}\cos x$.
c. Solve exactly on $[0, 2\pi]$: $\sin x + \sqrt{3}\cos x = 1$.

16. In a harbour the water height (in cm above a reference level) is modelled by

$$h(t) = 120 + 90\sin\left(\frac{2\pi}{12.4}(t-2)\right),$$

with t in hours after midnight.

a. Give the period and the times of the first high tide and the first low tide after midnight.
b. A ship needs a water height of more than 165 cm. Bereken exact, and then in hours and minutes, how long per tidal period the ship can enter.

17.

a. Prove that $\cos 3x = 4\cos^3 x - 3\cos x$. *Hint:* $3x = 2x + x$.
b. Show that $c = \cos\frac{\pi}{9}$ satisfies $8c^3 - 6c - 1 = 0$.

18. Two observers A and B stand on a straight coast, 500 m apart. A boat C is at sea. The angles $\angle CAB = 52°$ and $\angle CBA = 71°$ are measured. Calculate the distance from the boat to the coast, rounded to whole metres.

19. In triangle ABC: $b = 6$, $c = 10$ and $\alpha = 150°$.

a. Bereken exact the area.
b. Bereken exact a^2, and give a to two decimals.

20. Given $f(x) = \sin^2 x$.

a. Toon aan that $f(x) = \frac{1}{2} - \frac{1}{2}\cos 2x$.
b. Give the equilibrium, amplitude and period of the graph of f.

21. Solve exactly on $[0, 2\pi]$: $2\cos^2 x - 3\cos x + 1 = 0$.

Challenge

22. A second proof of the difference formula. Let $0 < \beta < \alpha < \beta + \pi$ and let $A = (\cos\alpha, \sin\alpha)$, $B = (\cos\beta, \sin\beta)$ be points on the unit circle with centre O.

a. Use the cosine rule in triangle OAB to express AB^2 in $\cos(\alpha - \beta)$.
b. Use coordinates to express AB^2 in $\cos\alpha, \cos\beta, \sin\alpha, \sin\beta$, and conclude $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$.
c. Why does this proof cover fewer cases than the rotation proof in the notes?

23. Triangle ABC has circumscribed circle with centre M and radius R. Prove that $\dfrac{a}{\sin\alpha} = 2R$. *Hint:* draw the diameter through B, with other endpoint D, and use the inscribed angle theorem (*stelling van de constante hoek*) to compare $\angle BDC$ with α. Treat the case where α is acute; then explain what changes if α is obtuse.

24. Let $\theta = \frac{\pi}{10}$ ($18°$).

a. Explain why $\sin 2\theta = \cos 3\theta$.
b. Use the double-angle formula and problem 17a to derive $4\sin^2\theta + 2\sin\theta - 1 = 0$.
c. Conclude that $\sin\frac{\pi}{10} = \dfrac{\sqrt{5}-1}{4}$.

25.

a. Prove that $\tan(\alpha+\beta) = \dfrac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$ (when all tangents are defined and the denominator is not 0).
b. Prove that $\arctan\frac{1}{2} + \arctan\frac{1}{3} = \frac{\pi}{4}$. (arctan is the inverse of tan on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.)

26. Heron's formula. A triangle has sides a, b, c and semi-perimeter $s = \frac{a+b+c}{2}$. Prove that its area is

$$\sqrt{s(s-a)(s-b)(s-c)}.$$

Hint: start from area $= \frac{1}{2}ab\sin\gamma$, write $\sin^2\gamma = 1-\cos^2\gamma = (1-\cos\gamma)(1+\cos\gamma)$, and substitute the cosine rule.